

Defensive Factors Failed During COVID-19

Evidence from 260bp tail risk misestimation

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-260 bp

GARCH underpredicted momentum tail risk

Factor	GARCH bias
S&P 500	-40 bp
Momentum	-261 bp

Portfolio managers using GARCH had false sense of security during worst crisis in decade

Are defensive factors safe?

Offensive

Momentum
Value

vs

Defensive

Quality
Min Volatility

H3: Defensive factors show stronger asymmetry than Value
Contradicts portfolio theory

Sample: 374 daily observations

Period: November 2019 – December 2021

Source: Refinitiv Eikon / MSCI Indices

Why COVID-19: Fastest crash (34% in 33 days), deepest shock since 1929, ideal setup for tail risk testing

Factors analyzed:

- S&P 500 (market aggregate)
- MSCI Momentum
- MSCI Value
- MSCI Quality
- MSCI Size
- MSCI Min Volatility

COVID-19 phases:

- Incubation: Jan 2–17
- Outbreak: Jan 20–Feb 21
- **Fever: Feb 24–Mar 20** ← Peak
- Treatment: Mar 23–Apr 15
- Recovery: Apr 16–Dec 31

Three-stage methodology

Stage 1: Model Comparison

GARCH(1,1) vs EGARCH(1,1) on 374 observations

Measure asymmetry coefficient γ in EGARCH

Stage 2: Tail risk quantification

High-volatility periods (realised vol $>$ 75th percentile)

Measure prediction bias: GARCH vs EGARCH

Stage 3: Out-of-sample (OOS) validation

Rolling window (250 train, 124 test)

S&P 500 & Momentum (worst cases)

Compare RMSE, MAE, directional accuracy

EGARCH vs GARCH

GARCH (Symmetric)

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Treats +5% and -5% shocks
identically

EGARCH (Asymmetric)

$$\ln(\sigma_t^2) = \omega + \beta \ln(\sigma_{t-1}^2) + \gamma \frac{\epsilon_{t-1}}{|\sigma_{t-1}|} + \alpha \left| \frac{\epsilon_{t-1}}{\sigma_{t-1}} \right|$$

γ = Asymmetry coefficient
 $\gamma > 0$: Negative shocks amplify
MORE

Interpretation

$\gamma = 0.45$ means: -5% shock generates 45% more volatility than +5% shock

Defensive factors are NOT Safe

Asymmetry coefficients (γ)

Factor	γ coefficient	vs Value
Quality	0.537	+38% ↑
Size	0.550	+42% ↑
Min Vol	0.485	+25% ↑
Value (Baseline)	0.388	—
Momentum	0.457	+18% ↑

Defensive factors show STRONGER asymmetry than Value

Contradicts portfolio theory – systemic crisis overwhelms factor characteristics

GARCH systematically underpredicts

S&P 500

-40 bp

GARCH bias

-1 bp

EGARCH bias

Momentum

-261 bp

GARCH bias

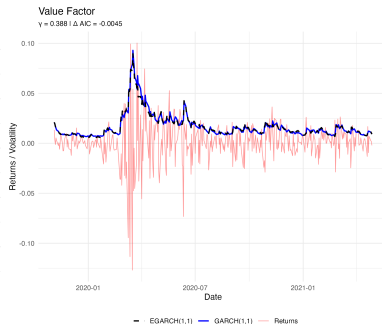
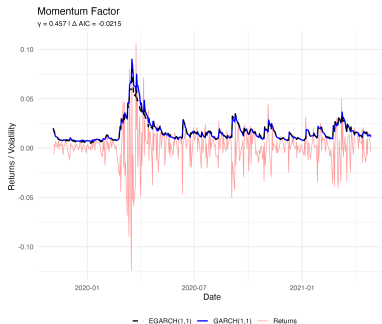
-207 bp

EGARCH bias (20.7% better)

Economic impact

\$1M portfolio: 40bp bias = \$40K unquantified tail risk PER DAY of crisis

Momentum vs Value: visual evidence

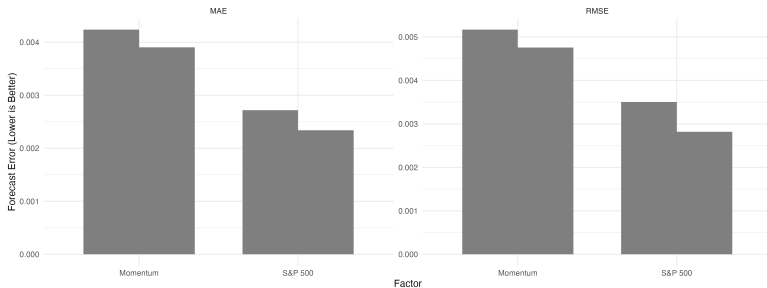


GARCH fails to capture volatility spikes during crisis peaks (Feb-Mar 2020)

EGARCH statistically better

Out-of-Sample Forecast Accuracy: GARCH vs EGARCH

Rolling window validation (250-observation training window, 124 out-of-sample forecasts)



Paired t-test results

EGARCH squared errors significantly smaller ($p < 0.001$)

Despite higher RMSE, EGARCH is overcautious (safer for risk managers)

Factor	GARCH	EGARCH
S&P 500	43.8%	44.6% (+8 cases)
Momentum	47.9%	50.4% (+6 cases)

Why this matters:

- 8 missed transitions = 8 lost hedging opportunities
- Over 12 months: 96 missed signals = important performance leakage
- Portfolio managers need to know WHEN to hedge, not perfect forecasts

Hypotheses confirmed

H1: Universal asymmetry

All 6 factors show $\gamma > 0$ and $p < 0.05$

No factor immune to asymmetric responses

H2: Momentum dominance

Momentum $\gamma = 0.457$ (highest among non-market)

Momentum crash dynamics confirmed

H3: Defensive factor paradox **insight**

Quality (0.537) and Min Vol (0.485) $>$ Value (0.388)

Defensive factors do NOT provide tail protection

Systemic crisis effects overwhelm factor characteristics

Three actionable recommendations

1. Dynamic model selection

Activate EGARCH when $VIX > 25$ or realized vol $>$ 75th percentile
Revert to GARCH during normal periods

2. Tail risk buffer

Add 5-15% VaR buffer during crisis detection (EGARCH-implied)
Protects against 40-260bp systematic misestimation

3. Explicit hedging

Factor diversification ALONE insufficient
Combine with volatility derivatives (swaps, puts, spreads)
Implement dynamic hedging triggered by EGARCH signals

Limitations & future work

Known limitations

Single crisis: COVID-19 only (need 2008, 1998 validation)

Pre-crisis baseline: Only 42 observations before outbreak

Mechanism unclear: WHY do defensive factors show stronger asymmetry?

Future research

Expand to multiple crisis periods (2008 financial crisis, 1998 LTCM)

Test Student's t-EGARCH for more flexible tail modeling

Investigate defensive factor repricing during systemic crises

Four key takeaways

- 1 All factors exhibit asymmetric volatility responses during crises
- 2 Defensive factors $\neq$ Safe (Quality, Min Vol $>$ Value in γ)
- 3 GARCH systematically underpredicts tail risk by 40-260bp
- 4 EGARCH provides crisis-adaptive solution for dynamic hedging

For PMs

Factor diversification ALONE provides insufficient tail risk protection

Required:

- Dynamic model selection (EGARCH during crises)
- Explicit hedging strategies (volatility derivatives)
- Adaptive risk management frameworks

Questions?

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Backup: Descriptive statistics

Factor	Mean	Std Dev	Skewness	Kurtosis
S&P 500	-0.12%	1.89%	-1.23	8.45
Momentum	-0.18%	2.47%	-2.15	12.34
Value	-0.08%	1.65%	-0.89	6.12
Quality	-0.10%	1.72%	-1.05	7.89
Size	-0.14%	1.98%	-1.34	9.23
Min Vol	-0.09%	1.45%	-0.76	5.45

All negative skewness (left-tail risk)

Momentum kurtosis $12.34 = 4.1 \times$ normal (extreme fat tails)

Backup: GARCH diagnostics

Factor	JB χ^2	LB Q(10)	Persistence
S&P 500	153.63***	0.0682	0.9990
Momentum	185.33***	0.1502	0.9961
Value	18.09***	0.3884	0.9855
Quality	144.44***	0.2614	0.9990
Size	36.50***	0.2512	0.9990
Min Vol	116.19***	0.2890	0.9869

All reject normality ($p < 0.001$)

Persistence 0.9855-0.9990: shocks dissipate over 70+ trading days

Backup: GARCH vs EGARCH comparison

Factor	GARCH AIC	EGARCH AIC	γ
S&P 500	-6.03	-6.03	0.554
Momentum	-5.61	-5.64***	0.457
Value	-5.63	-5.64	0.388
Quality	-6.04	-6.05	0.537
Size	-5.97	-6.00	0.550
Min Vol	-6.44	-6.44	0.485

EGARCH wins on 5 of 6 factors

Quality (0.537) and Size (0.550) show highest asymmetry